

The traditional virtual machine approach

The traditional virtual machine approach involves the creation of a virtualised environment designed to mimic the capabilities of a physical computer. This approach is widely used in enterprise environments to consolidate physical infrastructure and improve efficiency. It involves the following steps.

Hypervisor installation: A hypervisor is a layer of software that runs directly on the host computer that creates and manages virtual machines. It allocates resources from the physical hardware, such as CPU, memory, and storage, to each virtual machine.

Virtual machine creation: Once the hypervisor is installed, virtual machines can be created based on specific requirements including operating system, memory, and storage.

Operating system installation: Once the virtual machine is created, it must be configured, and an operating system must be installed in it. This involves setting up user accounts, network configurations, and other system settings.

Application installation: After the operating system has been installed, applications can be installed as per the requirements. As these consume system resources, it is important to allocate adequate resources to the virtual machines.

Maintenance: Virtual machines require regular maintenance, including security updates, software patches, and performance tuning. This can be carried out using tools such as configuration management tools, monitoring tools etc.

In the traditional virtual machine approach, each virtual machine runs separately and does not share resources with other virtual machines. This approach has its benefits, such as better isolation between applications, but it has some disadvantages too, such as less efficient use of physical resources.

Differences between virtual machines and containers

Virtual machines and containers are two different technologies used to create isolated environments in computing, but with distinct differences.

Architecture: Virtual machines run on a hypervisor layer that sits on top of the host operating system. Each virtual machine runs its own guest operating system, which is completely isolated from the host system. Containers, on the other hand, use a shared kernel architecture, where multiple virtualised environments share the same operating system kernel as the host operating system.

Resource consumption: Virtual machines require more resources than containers. Each virtual machine requires its own operating system, which means that it

requires more CPU, memory, and disk space. Containers, on the other hand, have a small overhead compared to virtual machines. The shared kernel architecture of containers takes up less memory and CPU, which means that more containers can be deployed on a single host than virtual machines.

Speed and performance: Virtual machines provide an isolated environment, which means that they are slower to start up than containers. They also have more overhead in communication between the guest and host operating systems, which can affect their performance. Containers, on the other hand, can start up quickly and have almost no overhead in communication between the container and the host operating system, which makes them faster and more performant.

Security: Virtual machines are more secure than containers because they provide a complete sandboxed environment, which means that they are completely isolated from the host system. Containers, on the other hand, run on the host operating system, which means that they have a smaller attack surface and are more vulnerable to attacks.

Application compatibility: Virtual machines are more versatile than containers because they can run any operating system. Containers, on the other hand, require the same kernel as the host operating system, which means that they are limited in the types of applications that they can run.

In conclusion, virtual machines are better suited for running different types of operating systems, whereas containers are better suited for running lightweight and scalable applications. Each has its own advantages and disadvantages, and choosing one over the other will depend on the specific use case.

Why are Linux containers gaining popularity over virtual machines?

Linux containers are becoming increasingly popular in enterprise computing due to their efficiency and portability. There are other reasons too.

Resource efficiency: Linux containers share the same kernel as the host operating system, meaning they require fewer resources to run than virtual machines. This makes them ideal for environments where resources are limited.

Faster startup times: Containers have faster startup times than virtual machines, making them ideal for applications that require quick scaling and deployment.

Portability: They can run on any system that supports the container runtime, making them highly portable across different architectures and operating systems.

Better utilisation of hardware: Containers distribute computing power more efficiently than virtual machines,